

Task 4: Smart Attendance Simulator

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Task Title

Smart Attendance Simulator

Objective

The objective of this task is to develop a Python-based attendance management system that reads student data from a CSV file, accepts user input, updates attendance records, and saves the updated information automatically.

Tools and Technologies Used

- Python
- Pandas
- Google Colab
- CSV Files
- GitHub

Introduction

Attendance management is a common task in schools, colleges, and organizations. Manual attendance recording can be time-consuming and prone to errors. This project automates the attendance process using Python and CSV files.

Procedure

Step 1: Create Student Data

A CSV file named students.csv was created containing student names.

Step 2: Load CSV File

The CSV file was loaded into Python using the Pandas library.

Step 3: Add Attendance Column

An Attendance column was created and initialized with the value "Absent" for all students.

Step 4: Accept User Input

The program asked the user to enter a student name.

Step 5: Match Student Name

The entered name was compared with names stored in the CSV file using case-insensitive matching.

Step 6: Mark Attendance

If a match was found, the student's attendance status was updated to "Present".

Step 7: Save Updated Data

The updated attendance records were saved into a new CSV file named `updated_attendance.csv`.

Code Implementation

```
import pandas as pd

df = pd.read_csv("students.csv")

df["Attendance"] = "Absent"

name = input("Enter student name: ").strip().lower()

for i in range(len(df)):
    if df.loc[i, "Name"].lower() == name:
        df.loc[i, "Attendance"] = "Present"

df.to_csv("updated_attendance.csv", index=False)

print(df)
```

Results

Original Student Data

```
import pandas as pd

df = pd.read_csv("students.csv")
print(df)
```

	Name
0	Vaishnavi
1	Rahul
2	Priya
3	Amit
4	Sneha

Attendance Marking Output

```
import pandas as pd

df = pd.read_csv("students.csv")

df["Attendance"] = "Absent"

name = input("Enter student name: ").strip().lower()

for i in range(len(df)):
    if df.loc[i, "Name"].lower() == name:
        df.loc[i, "Attendance"] = "Present"

df.to_csv("updated_attendance.csv", index=False)

print(df)
```

Enter student name: Vaishnavi

	Name	Attendance
0	Vaishnavi	Present
1	Rahul	Absent
2	Priya	Absent
3	Amit	Absent
4	Sneha	Absent

Updated Attendance File

```
updated = pd.read_csv("updated_attendance.csv")
print(updated)
```

	Name	Attendance
0	Vaishnavi	Present
1	Rahul	Absent
2	Priya	Absent
3	Amit	Absent
4	Sneha	Absent

Observations

- Student data was successfully loaded from a CSV file.
- Attendance was updated based on user input.
- Case-insensitive matching improved usability.
- The updated data was saved automatically in a new CSV file.

Learning Outcomes

- Learned CSV file handling using Pandas.
- Learned how to read and write CSV files.
- Learned user input processing.

- Learned data manipulation using DataFrames.
- Gained practical experience in attendance management automation.

Conclusion

The Smart Attendance Simulator successfully automated the process of recording student attendance. The system reads student records, updates attendance based on user input, and saves the modified data into a new CSV file. This task provided hands-on experience with Python, Pandas, and file handling techniques.